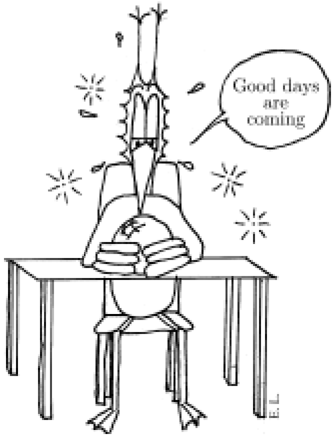
ΣΥΣΤΗΜΑΤΑ ΑΝΑΜΟΝΗΣ
(Queuing Systems)

Forecasting - Filters



Book from where to read:

Performance Evaluation of Computer and Communication Systems, **Jean-Yves Le Boudec, EPFL Press, Distributed by CRC Press**

Contents

1. Differencing Filters

- We have seen that changing the scale of the data may be important for obtaining a good model, e.g., using a logarithmic scale.

- *Filters perform a pre-processing of data*.: The idea is that the filter may remove deterministic patterns and it may be simpler to forecast the filtered data.

- *Filter Definition*: A filter (in full, discrete-time causal filter) is a mapping from the set of finite-length time series to the same set. (By convention, we consider that a filter keeps the length of the timeseries unchanged.)

- Further, a filter has to be linear, time-invariant and causal. The latter means that output of the filter up to time depends only on the input up to time.

Differencing Filter Δ_1

Differencing filter Δ_1= discrete-time derivative

$Y = (Y_1, ..., Y_t) \to \Delta_1 Y = (Y_1, Y_2 - Y_1, ..., Y_t - Y_{t-1})$
$\Delta_1 Y = X$ means that X has same length as Y and $X_t = Y_t - Y_{t-1}$
with the convention that $Y_t = 0$ whenever $t \leq 0$

Δ_1 is a filter, thus it is linear, and
$\Delta_1(Y + Z) = \Delta_1 Y + \Delta_1 Z$
If $Y_t = Z_t + at$ then $(\Delta_1 Y)_t = (\Delta_1 Z)_t + a$
Thus, Δ_1 removes linear trends.
Repetitive application of Δ_1 removes polynomial trends.

De-seasonalizing Filters

De-seasonalizing R_S: (sum of the last s values)

$R_S Y = X$ where X has the same length as Y and
$(R_S Y)_t = Y_{t-s+1} + ... + Y_{t-1} + Y_t$
with the convention that $Y_t = 0$ whenever $t \leq 0$

If Y_t is periodic of period s then $R_S Y$ is constant
In other words, $R_S Y$ removes the periodic components

Differencing Δ_s:
$\Delta_s Y = X$ where X has the same length as Y and $X_t = Y_t - Y_{t-s}$
with the convention that $Y_t = 0$ whenever $t \leq 0$

De-seasonalizing Filters

$$\Delta_s = R_s \Delta_1$$

this means that if $Y \overset{\Delta_1}{\rightarrow} Z \overset{R_s}{\rightarrow} X$ and $Y \overset{\Delta_s}{\rightarrow} X'$ then $X = X'$

Proof:

$$Z_t = Y_t - Y_{t-1}$$
$$X_t = Z_t + \cdots + Z_{t-s+1} = Y_t - Y_{t-1} + Y_{t-1} - Y_{t-2} \ldots + Y_{t-s+1} - Y_{t-s}$$
$$= Y_t - Y_{t-s}$$

Questions

Which matrix is the representation of Δ_1 (over time series of length n) ?

$$A.\ \ A = \begin{pmatrix} 1 & 0 & \dots & \dots & \dots & 0 \\ -1 & 1 & 0 & \dots & \dots & 0 \\ 0 & -1 & 1 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & -1 & 1 & 0 \\ 0 & 0 & 0 & \dots & -1 & 1 \end{pmatrix}$$

$$B.\ \ B = \begin{pmatrix} 1 & -1 & \dots & \dots & \dots & 0 \\ 0 & 1 & -1 & \dots & \dots & 0 \\ 0 & 0 & 1 & -1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & 1 & -1 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{pmatrix}$$

C. None of these

D. I don't know

Which matrix is the representation of R_4 (over time series of length $n = 6$) ?

$$A.\ \ A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$B.\ \ B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

C. None of these

D. I don't know

Say what is true

A. A

B. B

C. C

D. A,B

E. A,C

F. B,C

G. All

H. None

I. I don't know

A. $R_s\Delta_1 = \Delta_s$
B. $\Delta_1 R_s = \Delta_s$
C. $\Delta_1\Delta_1 = \Delta_2$

Questions – Response 1

Answer A: $(\Delta_1 X)_t = X_t - X_{t-1}$

$$\begin{pmatrix} X_1 \\ X_2 - X_1 \\ X_3 - X_2 \\ \ldots \\ X_{n-1} - X_{n-2} \\ X_n - X_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \ldots & \ldots & \ldots & 0 \\ -1 & 1 & 0 & \ldots & \ldots & 0 \\ 0 & -1 & 1 & 0 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & \ldots & -1 & 1 & 0 \\ 0 & 0 & 0 & \ldots & -1 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ \ldots \\ X_{n-1} \\ X_n \end{pmatrix}$$

Questions – Response 2

Answer B: $(R_4X)_t = X_{t-3} + X_{t-2} + X_{t-1} + X_t$

$$\begin{pmatrix} X_1 \\ X_1 + X_2 \\ X_1 + X_2 + X_3 \\ X_1 + X_2 + X_3 + X_4 \\ X_2 + X_3 + X_4 + X_5 \\ X_3 + X_4 + X_5 + X_6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{pmatrix}$$

Questions – Response 3

A is true as we saw earlier

B is true. We prove by a direct computation as we did earlier, or what we can use the property that *any two filters commute,* i.e. the order in which a succession of filters is applied does not matter: $FG = GF$ for any two filters F, G

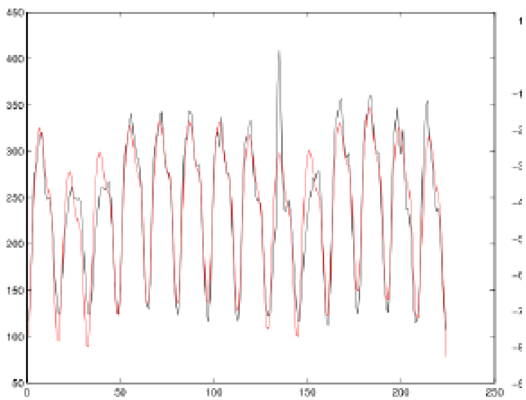
C is false. Let us compute $X = \Delta_1 \Delta_1 Y$. Let $Z = \Delta_1 Y$, so that

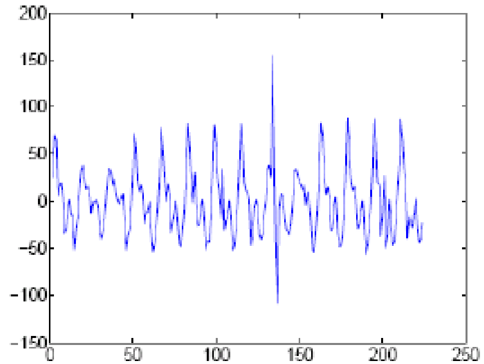
$$\begin{aligned} X_t &= Z_t - Z_{t-1} \\ &= (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) = Y_t - 2\,Y_{t-1} + Y_{t-2} \end{aligned}$$

which is not equal to $Y_t - Y_{t-2}$

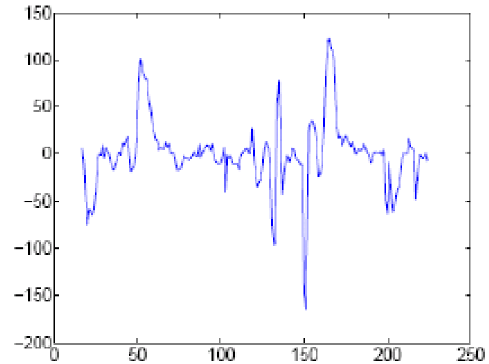
$\Delta_1 \Delta_1$, also noted $(\Delta_1)^2$ is the discrete-time second derivative.

Example

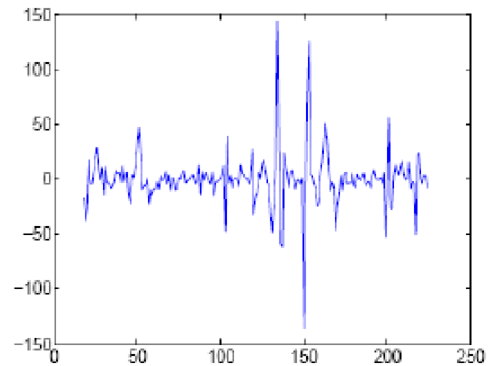




(a) Differencing at Lag 1



(b) Differencing at Lag 16



(c) Differencing at Lags 1 and 16

EXAMPLE 5.3: INTERNET TRAFFIC. In Figure 5.7 we apply the differencing filter Δ_1 to the time series in Example 5.1 and obtain a strong seasonal component with period $s = 16$. We then apply the de-seasonalizing filter R_{16}; this is the same as applying Δ_{16} to the original data. The result does not appear to be stationary; an additional application of Δ_1 is thus performed.

2. Filters For Dummies

To make predictions with filters, we need first to see how filters work……and this is what we learn in this section.

D.1.1 Backshift Operator

We consider data sequences of finite, but arbitrary length and call $\mathcal{S}$ the set of all such sequences (i.e. $\mathcal{S} = \bigcup_{n=1}^{\infty} \mathbb{R}^n$). We denote with length$(X)$ the number of elements in the sequence X.

The *backshift* operator is the mapping B from $\mathcal{S}$ to itself defined by:

$$\begin{aligned} \text{length}(BX) &= \text{length}(X) \\ (BX)_1 &= 0 \\ (BX)_t &= X_{t-1} \qquad t = 2, ..., \text{length}(X) \end{aligned}$$

We usually view a sequence $X \in \mathcal{S}$ as a column vector, so that we can write:

$$B \begin{pmatrix} X_1 \\ X_2 \\ \dots \\ X_n \end{pmatrix} = \begin{pmatrix} 0 \\ X_1 \\ \dots \\ X_{n-1} \end{pmatrix} \tag{D.1}$$

when length$(X) = n$.

Backshift Operator in Matrix Form

$(BX)_t = X_{t-1}$

B

$$\begin{pmatrix} 0 \\ X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{pmatrix}$$

$(B^2X)_t = X_{t-2}$

B^2

$$\begin{pmatrix} 0 \\ 0 \\ X_1 \\ X_2 \\ X_3 \\ X_4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{pmatrix}$$

Filter Definition

DEFINITION D.1. *A filter (also called "causal filter", or "realizable filter") is any mapping, say F, from $\mathcal{S}$ to itself that has the following properties.*

1. *A sequence of length n is mapped to a sequence of same length.*
2. *There exists an infinite sequence of numbers h_m, $m = 0, 1, 2, \ldots$ (called the filter's impulse response) such that for any $X \in \mathcal{S}$*

$$(FX)_t = h_0 X_t + h_1 X_{t-1} + \ldots + h_{t-1} X_1 \qquad t = 1, \ldots, \mathit{length}(X) \tag{D.4}$$

In matrix form, if we know that length$(X) \leq n$ we can write Eq.(D.4) as

$$FX = \begin{pmatrix} h_0 & 0 & \cdots & 0 & 0 \\ h_1 & h_0 & & \vdots & \vdots \\ h_2 & h_1 & \ddots & & \\ \vdots & \vdots & \ddots & h_0 & 0 \\ h_{n-1} & h_{n-2} & \cdots & h_1 & h_0 \end{pmatrix} X \tag{D.6}$$

Operator Notation

Example: $(FX)_t = 3X_t - 2X_{t-1} + X_{t-2}$

Impulse response: $h_0 = 3, h_1 = -2, h_2 = 1, h_k = 0, k \geq 3$

Matrix form:
$$FX = \begin{pmatrix} 3 & 0 & 0 & 0 & 0 & 0 \\ -2 & 3 & 0 & 0 & 0 & 0 \\ 1 & -2 & 3 & 0 & 0 & 0 \\ 0 & 1 & -2 & 3 & 0 & 0 \\ 0 & 0 & 2 & -2 & 3 & 0 \\ 0 & 0 & 0 & 1 & -2 & 3 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{pmatrix}$$

$$M = 3\,\mathrm{Id} - 2\,B + B^2$$

In operator notation we write
$$F = 3 - 2B + B^2$$

where B is the backshift filter.

Filters

Operator notation of F	Input-Output Equation $Y = FX$	Impulse Response
1	$Y_t = X_t$	$(1,0,0,\dots)$
B	$Y_t = X_{t-1}$	$(0,1,0,0,\dots)$
$\Delta_1 = 1 - B$	$Y_t = X_t - X_{t-1}$	$(1,-1,0,0,\dots)$
$\Delta_s = 1 - B^s$	$Y_t = X_t - X_{t-s}$	$(1,0,\dots,0,-1,0,0,\dots)$
$R_s = 1 + B + \cdots + B^{s-1}$	$Y_t = X_t + \cdots + X_{t-s+1}$	$(1,\dots,1,0,0,\dots)$
$F = h_0 + h_1 B + h_2 B^2 + \cdots$	$Y_t = h_0 X_t + h_1 X_{t-1} + \cdots$	$(h_0, h_1, \dots)$
$\frac{1}{F}$ defined when $h_0 \neq 0$	$X_t = h_0 Y_t + h_1 Y_{t-1} + \cdots$ i.e. $Y_t = \frac{1}{h_0} X_t - \frac{h_1}{h_0} Y_{t-1} - \cdots$	

Impulse Response

Let F be a filter with impulse response $h_0, h_1, \ldots$

Thus $Y_t = h_0 X_t + h_1 X_{t-1} \ldots$

If the input X is the *impulse*

$$X = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \cdots \end{pmatrix}$$

then the output it

$$Y = \begin{pmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \\ h_4 \\ \cdots \end{pmatrix}$$

Inverse of a Filter

$$FX = \begin{pmatrix} h_0 & 0 & \cdots & 0 & 0 \\ h_1 & h_0 & & \vdots & \vdots \\ h_2 & h_1 & \ddots & & \\ \vdots & \vdots & \ddots & h_0 & 0 \\ h_{n-1} & h_{n-2} & \cdots & h_1 & h_0 \end{pmatrix} X$$

Filter F has an inverse if and only if $h_0 \neq 0$

Example: $\Delta_1 = 1 - B$ is invertible

Filter Calculus

FG means the operator composition (G followed by F):

$$Y \xrightarrow{G} Z \xrightarrow{F} X \text{ implies } Y \xrightarrow{FG} X$$

Filters commute: $FG = GF$ Magical !

$\frac{1}{F}$ means F^{-1}, the inverse of F, defined if $h_0 \neq 0$

$$\frac{G}{F} = G\,F^{-1} = F^{-1}G$$

$F + G$ means the algebraic sum: $[(F+G)X]_t = [FX]_t + [GX]_t$

Questions on Filters

Let $\boldsymbol{F}$ be the filter defined by $\boldsymbol{Y} = \boldsymbol{FX}$ with

$$\boldsymbol{Y_t = X_t - 3X_{t-1} + 2X_{t-2}}$$

Say what is true

A. $F = 1 - 3B + 2B^2$

B. The impulse response of F is $(1, -3,2,0,0 \dots)$

C. A and B

D. None

E. I don't know

Let $\boldsymbol{F}$ be the filter defined by $\boldsymbol{Y} = \boldsymbol{FX}$ with

$$\boldsymbol{Y_t - 3Y_{t-1} + 2Y_{t-2} = X_t}$$

Say what is true

A. $F = \frac{1}{1-3B+2\,B^2}$

B. The impulse response of F is $(1, -3,2,0,0 \dots)$

C. A and B

D. None

E. I don't know

Questions on Filters -Response

Let $\boldsymbol{F}$ be the filter defined by $\boldsymbol{Y} = \boldsymbol{F}\boldsymbol{X}$ with

$$\boldsymbol{Y_t = X_t - 3X_{t-1} + 2X_{t-2}}$$

Say what is true

A. $F = 1 - 3B + 2B^2$
B. The impulse response of F is $(1, -3, 2, 0, 0 \ldots)$
C. A and B
D. None
E. I don't know

Let $\boldsymbol{F}$ be the filter defined by $\boldsymbol{Y} = \boldsymbol{F}\boldsymbol{X}$ with

$$\boldsymbol{Y_t - 3Y_{t-1} + 2Y_{t-2} = X_t}$$

Say what is true

A. $F = \frac{1}{1 - 3B + 2B^2}$
B. The impulse response of F is $(1, -3, 2, 0, 0 \ldots)$
C. A and B
D. None
E. I don't know

ARMA Filters

Finite Impulse Response (FIR) also called Moving Average (MA) filter:

$h_k = 0$ for k that is large enough $\leftrightarrow F$ is polynomial in B
Example: Δ_1 is FIR but $\Delta_1^{-1} = 1 + B + B^2 + ...$ is not FIR
Generic Form of a FIR Filter: $F = h_0 + h_1 B + h_2 B^2 + ... + h_p B^p$

Auto-Regressive (AR) Filter is the inverse of a FIR filter

$1 + B + B^2 + ... = \frac{1}{1-B}$ is an AR filter
Generic AR Filter: $F = \frac{1}{h_0 + h_1 B + ... + h_p B^p}$ with $h_0 \neq 0$

ARMA Filter is *F/G* where *F, G* are FIR

Generic ARMA Filter: $F = \frac{f_0 + f_1 B + ... + F^q B^q}{h_0 + h_1 B + ... + h_p B^p}$

Implemented by Matlab’s filter() function

ARMA Filters

$Y = filter(P,\ Q,\ X)$ computes the output $Y = [y_1\ y_2\ y_3...\ y_n]$ of the filter where $P = [P_0\ P_1\ P_2...\ P_p]$ and $Q = [1\ Q_1\ Q_2...\ Q_q]$ are the filter coefficients and $X = [x_1\ x_2\ x_3...\ x_n]$ is the input. The filter is defined by the relation

$$y_k + Q_1 y_{k-1} + ... + Q_q y_{k-q} = P_0 x_k + P_1 x_{k-1} + ... + P_p x_{k-p}$$

where we set $x_i = 0$ and $y_i = 0$ when $i < 0$ or $i > n$.

Numerator polynomial: $P(\xi) = P_0 \xi^p + P_1 \xi^{p-1} + ... + P_p$
Denominator polynomial: $Q(\xi) = \xi^q + Q_1 \xi^{q-1} + ... + Q_q$

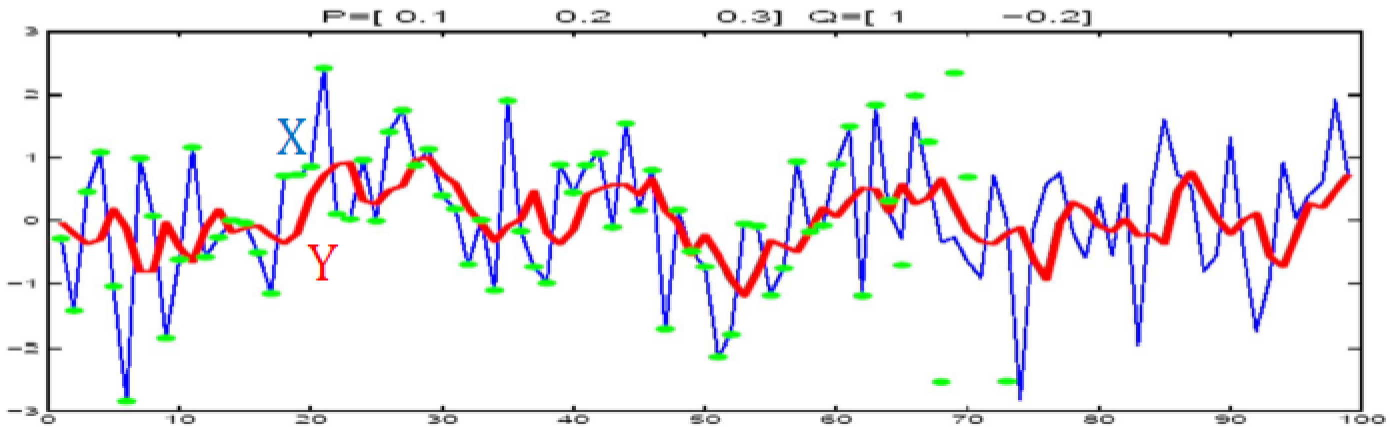
Filter is a mapping: $R^n \to R^n$

$$X \to Y = \frac{\sum_{i=0}^{p} P_i B^i}{Id + \sum_{j=1}^{q} Q_j B^j} X = \frac{P(B)}{Q(B)} X$$

Matlab	Operator Notation	Input-Output Equation
Y= filter ([0.1 0.2 0.3], [1 -0.2], X)	$Y = \frac{0.1+0.2B+0.3B^2}{1-0.2B} X$	$Y_t - 0.2Y_{t-1}$ $= 0.1X_t + 0.2X_{t-1}$ $+ 0.3\,X_{t-2}$

ARMA Filters Example (1)

A sample of $Y = \frac{0.1+0.2B+0.3B^2}{1-0.2B} X$



Blue X
Red Y
Green F^{-1}. that should estimate X

Q: how can we compute X back from Y ?

A: inverse the filter $X = \frac{1-0.2B}{0.1+0.2B+0.3B^2} Y$

The inverse of $Y = \text{filter}(P, Q, X)$ is $X = \text{filter}(Q, P, X)$ defined if first element of Q is $\neq 0$

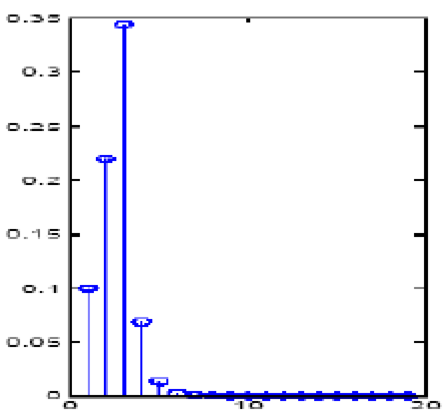
The result is shown with green dots; after $t = 60$ the results are incorrect. Why ?

ARMA Filters Example (2)

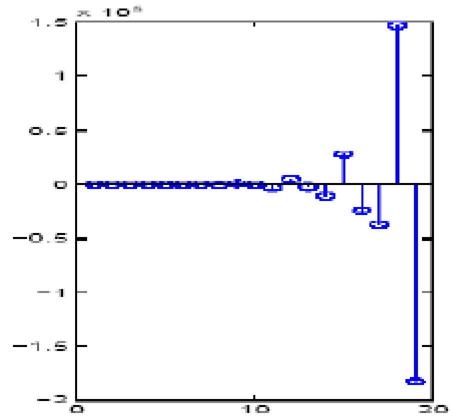
To understand what happens, let us compute the coefficients of these filters (i.e. their impulse responses)

It is obtained by $h = \text{filter}(P, Q, \text{imp})$ where $\text{imp} = [1\ 0\ 0\ \ ...]$ is called an impulse

Impulse response of $F = \frac{0.1+0.2B+0.3B^2}{1-0.2B}$



Impulse response of F^{-1}



The impulse of F^{-1} grows exponentially and becomes huge → numerical computation becomes impossible

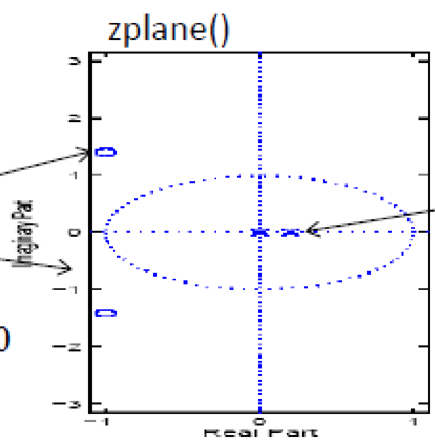
Filter BIBO Stability

A filter that is unstable usually causes numerical problems
(due to accumulation of rounding errors)

For an ARMA Filter: $F = \frac{f_0+f_1 B+...+F^q B^q}{1+h_1 B+...+h_p B^p}$, the poles are the p roots of the denominator polynomial $Q(\xi) = \xi^p + h_1 \xi^{p-1} + ... + Q_p$

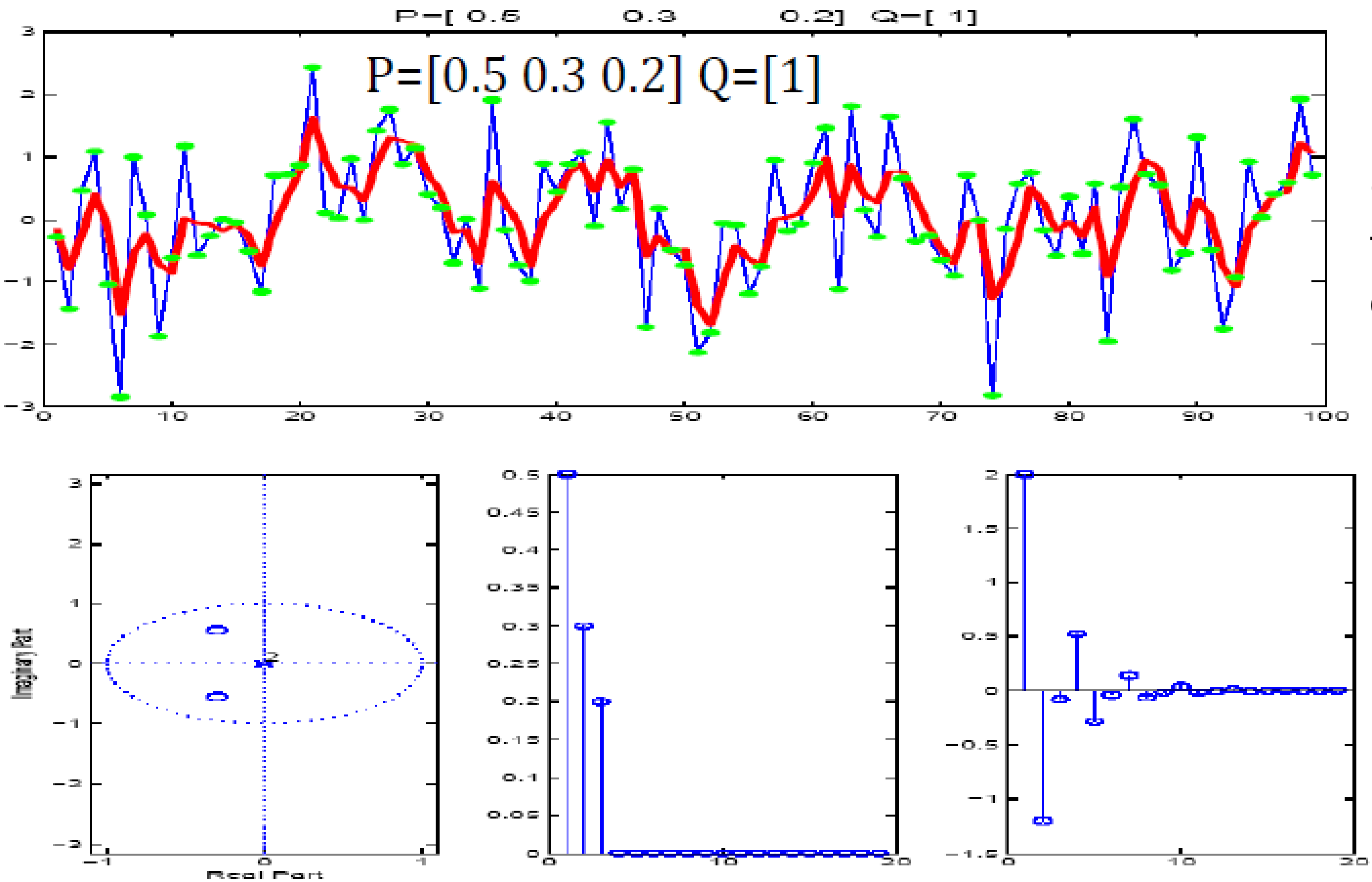
Definition of a stable filter: $p = 0$ i.e., it has no poles or all poles have module < 1

Zeroes of F
= Poles of F^{-1} =
Solutions of
$0.1\, z^2 + 0.2\, z + 0.3 = 0$



Pole of $F = \frac{0.1+0.2B+0.3B^2}{1-0.2B}$
Solution of $z - 0.2 = 0$

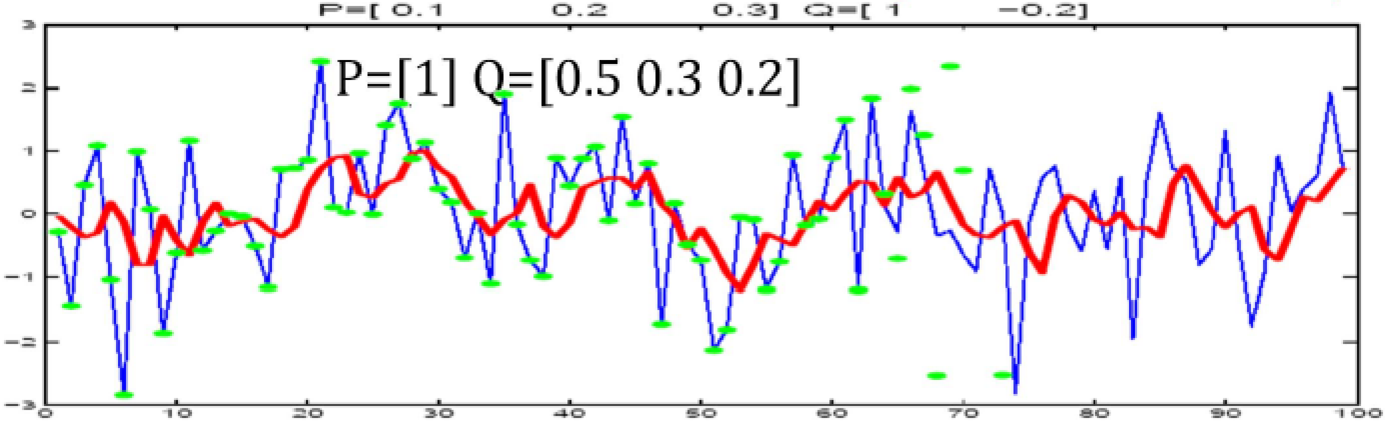
Example of a Stable Filter with a Stable Inverse



Blue X
Red Y
Green F^{-1} that should estimate X

Question on Filters

What is true about this filter $\boldsymbol{F}$ (where $Y = FX$)?



A. $0.5Y_t + 0.3\,Y_{t-1} + 0.2Y_{t-2} = X_t$

B. $Y_t = X_t - 0.5Y_t - 0.3\,Y_{t-1} - 0.2Y_{t-2}$

C. $Y_t = \dfrac{X_t}{0.5Y_t + 0.3Y_{t-1} + 0.2\,Y_{t-2}}$

D. A and B

E. A and C

F. B and C

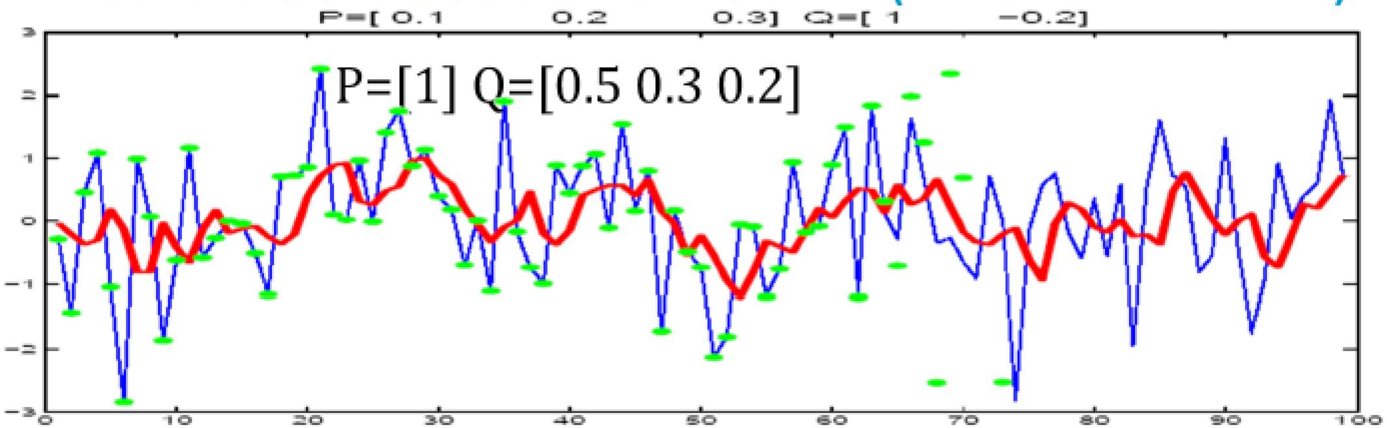
G. All

H. None

I. I don't know

Question on Filters - Response

What is true about this filter $\boldsymbol{F}$ (where $Y = FX$)?



A. $0.5Y_t + 0.3\, Y_{t-1} + 0.2Y_{t-2} = X_t$
B. $Y_t \; = \; X_t - 0.5Y_t - 0.3\, Y_{t-1} - 0.2Y_{t-2}$
C. $Y_t = \dfrac{X_t}{0.5Y_t + 0.3Y_{t-1} + 0.2\, Y_{t-2}}$
D. A and B
E. A and C
F. B and C
G. All
H. None
I. I don't know

Other Representations of Filters

Let $Y = FX$ with filter F with coefficients (given by impulse response) h_i

The MA(∞) representation of F is the standard input-output equation
$Y_t = h_0 X_t + h_1 X_{t-1} + ... + h_{t-1} X_1$

Assume that F is invertible and let h'_i be the impulse response of F^{-1} so that

$X_t = h'_0 Y_t + h'_1 Y_{t-1} + ... + h'_{t-1} Y_1$

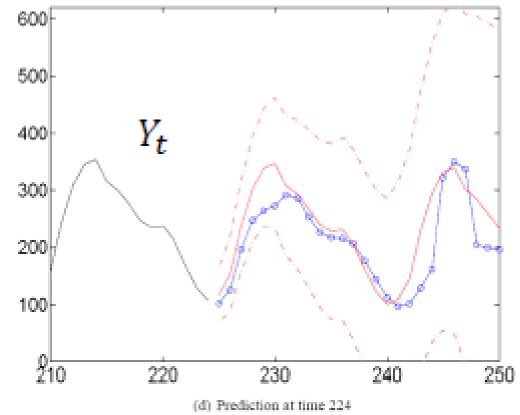
The AR(∞) representation of F is defined as

$Y_t = \frac{1}{h'_0} X_t - \frac{h'_1}{h'_0} Y_{t-1} - \frac{h'_2}{h'_0} Y_{t-2} - ... - \frac{h'_{t-1}}{h'_0} Y_1$

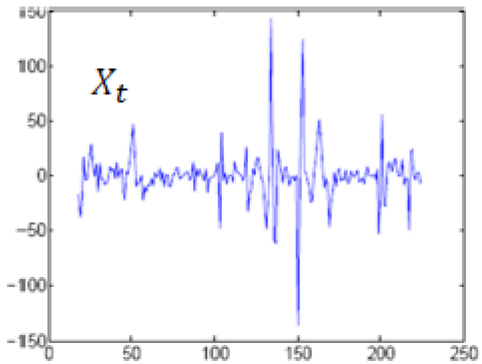
3. Predictions with Filters

How is this prediction done ?

Recall that $X = LY$ with
$L = \Delta_1 \Delta_{16}$ and we assume that X is i.i.d. $F()$
thus with the MA representation
$X_t = Y_t - Y_{t-1} - Y_{t-16} + Y_{t-17}$
or
$Y_t = X_t + Y_{t-1} + Y_{t-16} - Y_{t-17}$



(d) Prediction at time 224



(c) Differencing at Lags 1 and 16

To obtain a prediction at lag 1, i.e., at time $t+1$
we do the following:
Assume we know $Y_1, \ldots, Y_t$
Then $Y_{t+1} = X_{t+1} + Y_t + Y_{t-15} - Y_{t-16}$

Given the past up to time t, this is random with distribution F()

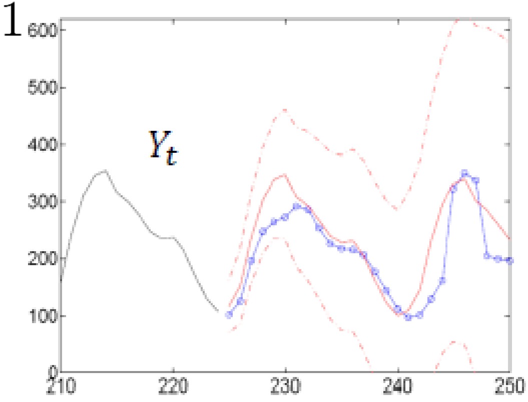
known

Point Prediction at lag 1

To obtain a prediction at lag 1, i.e., at time $t + 1$ we do the following:
Assume we know $Y_1, ..., Y_t$
Then $Y_{t+1} = X_{t+1} + Y_t + Y_{t-15} - Y_{t-16}$

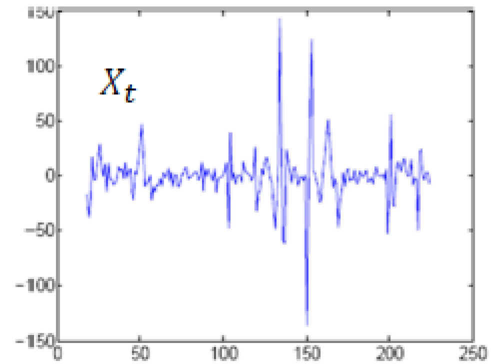
Given the past up to time t, this is random with distribution F()

known



(d) Prediction at time 224

Assume X is i.i.d. $F()$ with zero mean the mean of Y_{t+1} given the past up to time t is (point prediction)
$\hat{Y}_t(1) = Y_t + Y_{t-15} - Y_{t-16}$



(c) Differencing at Lags 1 and 16

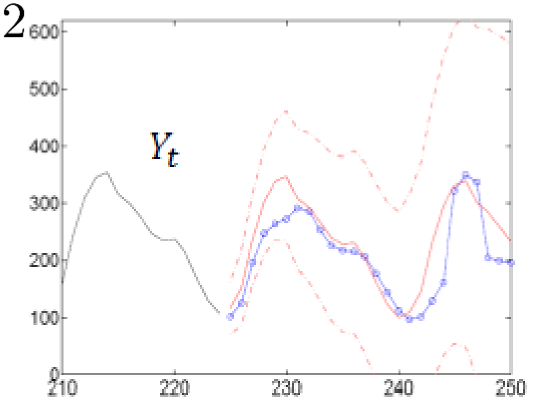
Point Predictions

To obtain a prediction at lag 2, i.e., at time $t+2$ we do the following:
Assume we know $Y_1, \ldots, Y_t$
Then $Y_{t+2} = X_{t+2} + Y_{t+1} + Y_{t-14} - Y_{t-15}$

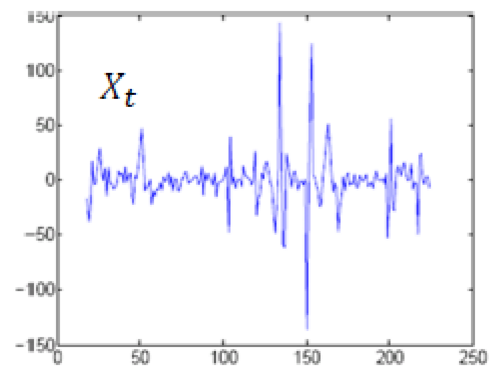
Given the past up to time t, the conditional expectation is 0 because F() has 0 mean

Given the past up to time t, the conditional expectation is $\hat{Y}_t(1)$

known



(d) Prediction at time 224



(c) Differencing at Lags 1 and 16

Therefore: (point prediction at lag 2)
$\hat{Y}_t(2) = \hat{Y}_t(1) + Y_{t-14} - Y_{t-15}$
At lag ℓ use the formula
$Y_{t+\ell} = X_{t+\ell} + Y_{t+\ell-1} + Y_{t+\ell-16} - Y_{t+\ell-17}$
in which you replace
Y_{t+s} with $\hat{Y}_t(s)$ for $s > 0$ and $X_{t+\ell}$ by 0, i.e., the mean of $F()$
For example $\hat{Y}_t(17) = \hat{Y}_t(16) + \hat{Y}_t(1) - Y_t$

Point Predictions

PROPOSITION 5.1. *Assume that $X = LY$ where L is a differencing or de-seasonalizing filter with impulse response $g_0 = 1, g_1, \ldots, g_q$. Assume that we are able to produce a point prediction $\hat{X}_t(\ell)$ for $X_{t+\ell}$ given that we have observed X_1 to X_t. For example, if the differenced data can be assumed to be iid with mean μ, then $\hat{X}_t(\ell) = \mu$.*

A point prediction for $Y_{t+\ell}$ can be obtained iteratively by:

$$
\begin{aligned}
\hat{Y}_t(\ell) &= \hat{X}_t(\ell) - g_1\hat{Y}_t(\ell-1) - \ldots - g_{\ell-1}\hat{Y}_t(1) - g_\ell y_t - \ldots \\
&\quad - g_q y_{t-q+\ell} \quad \text{for } 1 \le \ell \le q \qquad (5.14) \\
\hat{Y}_t(\ell) &= \hat{X}_t(\ell) - g_1\hat{Y}_t(\ell-1) - \ldots - g_q\hat{Y}_t(\ell-q) \text{ for } \ell > q \qquad (5.15)
\end{aligned}
$$

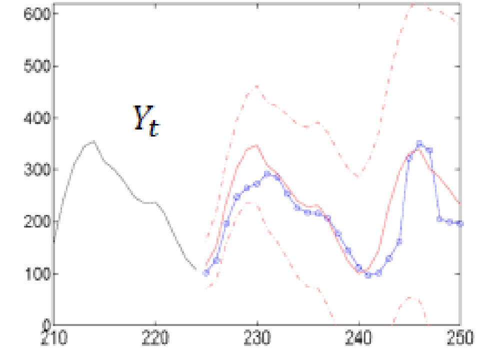
Use of the alternative representation the MA representation of the inverse Filter

Prediction at lag $\ell = 1$:
Assume we know $Y_1, \ldots, Y_t$

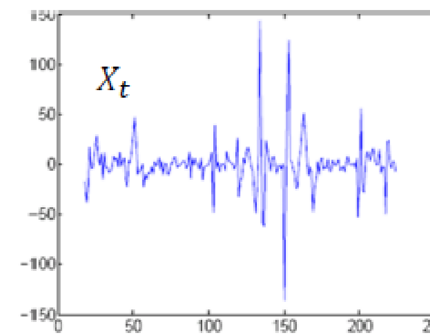
$$
\begin{aligned}
Y_{t+1} = X_{t+1} + X_t + \ldots + X_{t-14} \\
+ 2(X_{t-15} + X_{t-16} + \ldots + X_{t-30}) \\
+ 3(X_{t-31} + X_{t-32} + \ldots + X_{t-46}) + \ldots
\end{aligned}
$$

Given the past up to time t, this is random with distribution F()

known



(d) Prediction at time 224



(c) Differencing at Lags 1 and 16

Therefore

$$
\begin{aligned}
\hat{Y}_t(1) = X_t + \ldots + X_{t-14} \\
+ 2(X_{t-15} + X_{t-16} + \ldots + X_{t-30}) \\
+ 3(X_{t-31} + X_{t-32} + \ldots + X_{t-46}) + \ldots
\end{aligned}
$$

Note it would not be a good idea to use this formula to compute $\hat{Y}_t(1)$ because we accumulate a large number of errors – but it can be used to compute prediction intervals.

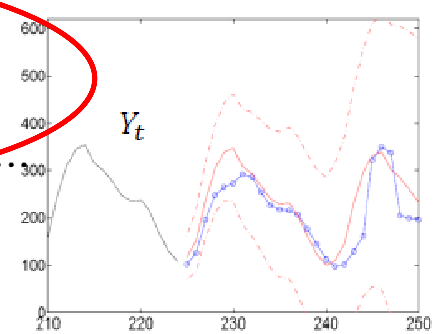
Computation of Prediction Intervals Example at lag 3

Assume we know $Y_1, ..., Y_t$

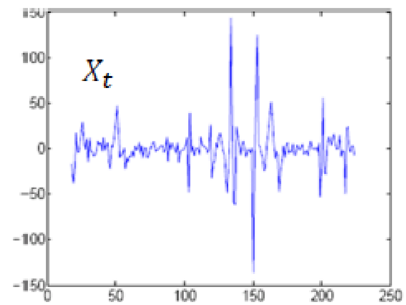
Prediction at lag $\ell = 3$: assume we know $Y_1, ..., Y_t$ Since the filter L is causal and invertible, knowing $Y_1, ..., Y_t$ is equivalent to knowing $X_1, ..., X_t$.

$$Y_{t+3} = X_{t+3} + X_{t+2} + X_{t+1} + X_t + ... + X_{t-12}$$
$$+ 2(X_{t-13} + X_{t-14} + ... + X_{t-28})$$
$$+ 3(X_{t-29} + X_{t-28} + ... + X_{t-44}) + ...$$

Known at time $t = \hat{Y}_t(3)$



(d) Prediction at time 224



(c) Differencing at Lags 1 and 16

Therefore: (innovation formula)
$$Y_{t+3} = X_{t+3} + X_{t+2} + X_{t+1} + \hat{Y}_t(3)$$

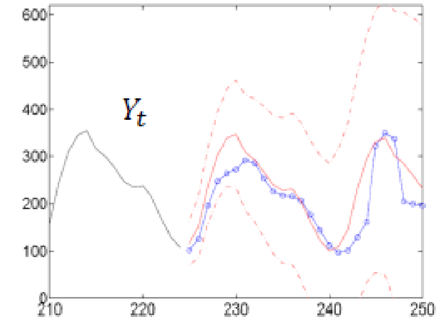
Computation of Prediction Intervals Example at lag 3

Therefore: (innovation formula)

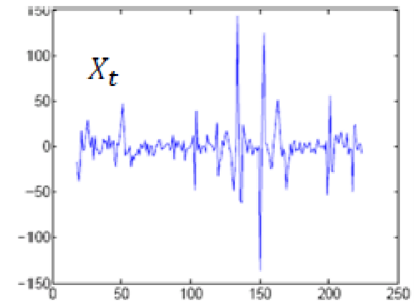
$$Y_{t+3} = X_{t+3} + X_{t+2} + X_{t+1} + \hat{Y}_t(3)$$

Given the past up to time t,
the distribution of Y_{t+3} is given by:
- a constant $\hat{Y}_t(3)$
- plus the sum of 3 independent random variables each with distribution $F()$ (the assumed distribution of X_t)

Example: assume that X_t follows iid $N(0, \sigma^2)$
then the distribution of $X_{t+3} + X_{t+2} + X_{t+1}$ is $N(0, 3\sigma^2)$
i.e., the distribution of Y_{t+3} given the past up to time t is normal with mean $\hat{Y}_t(3)$ and variance $3\sigma^2$.



(d) Prediction at time 224



(c) Differencing at Lags 1 and 16

Computation of Prediction Intervals

A 95%-prediction interval at lag 3 is...

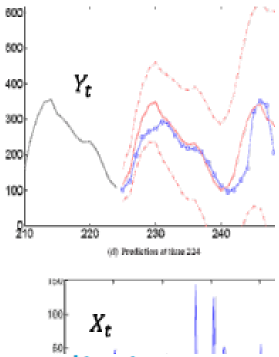
A. $Y_t(3) \pm 1.96\,\sigma$

B. $\hat{Y}_t(3) \pm 1.96 \times \sqrt{3}\,\sigma$

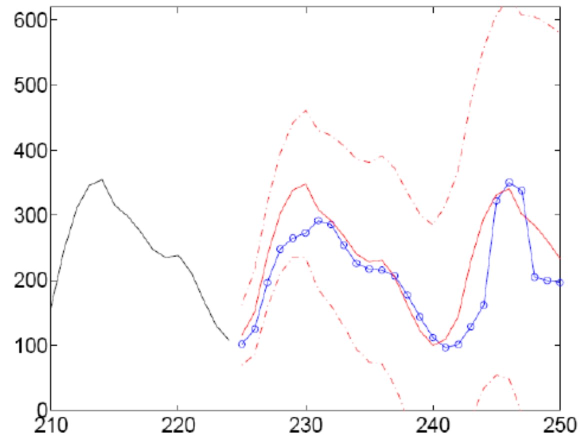
C. $\hat{Y}_t(3) \pm 1.96 \times 3\,\sigma$

D. $\hat{Y}_t(3) \pm 1.96 \times 3\frac{\sigma}{\sqrt{n}}$

E. None of the above



Prediction assuming differenced data is iid $\boldsymbol{N(0, \sigma^2)}$



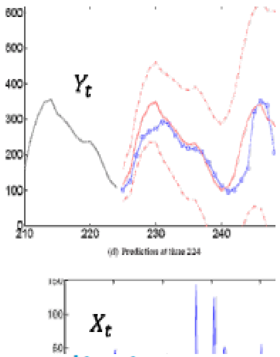
(d) Prediction at time 224

Figure 6.7: Differencing filters Δ_1 and Δ_{16} applied to Example 6.1 (first terms removed). The forecasts are made assuming the differenced data is iid gaussian with 0 mean. o = actual value of the future (not used for fitting the model).

Computation of Prediction Intervals - Response

A 95%-prediction interval at lag 3 is…

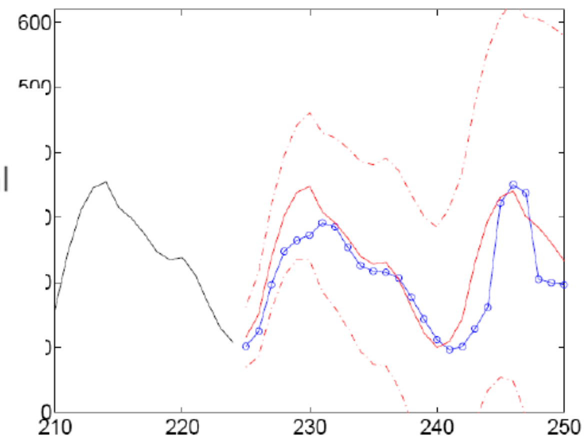
- *A.* $Y_t(3) \pm 1.96\,\sigma$
- *B.* $\hat{Y}_t(3) \pm 1.96 \times \sqrt{3}\,\sigma$
- *C.* $\hat{Y}_t(3) \pm 1.96 \times 3\,\sigma$
- *D.* $\hat{Y}_t(3) \pm 1.96 \times 3\frac{\sigma}{\sqrt{n}}$
- E. None of the above



Prediction assuming differenced data is iid $\boldsymbol{N(0,\sigma^2)}$

The distribution of Y_{t+3} given the past up to time t is normal with mean $\hat{Y}_t(3)$ and variance $3\,\sigma^2$, therefore with proba 95%, Y_{t+3} is in the interval $\hat{Y}_t(3) \pm 1.96 \times \sqrt{3}\,\sigma$

Answer B



(d) Prediction at time 224

Figure 6.7: Differencing filters Δ_1 and Δ_{16} applied to Example 6.1 (first terms removed). The forecasts are made assuming the differenced data is iid gaussian with 0 mean. o = actual value of the future (not used for fitting the model).

Computation of Prediction Intervals

PROPOSITION 5.2. *Assume that the differenced data is iid gaussian. i.e.* $X_t = (LY)_t \sim$ *iid* $N(\mu, \sigma^2)$. *The conditional distribution of* $Y_{t+\ell}$ *given that* $Y_1 = y_1, ..., Y_t = y_t$ *is gaussian with mean* $\hat{Y}_t(\ell)$ *obtained from Eq.(5.14) and variance*

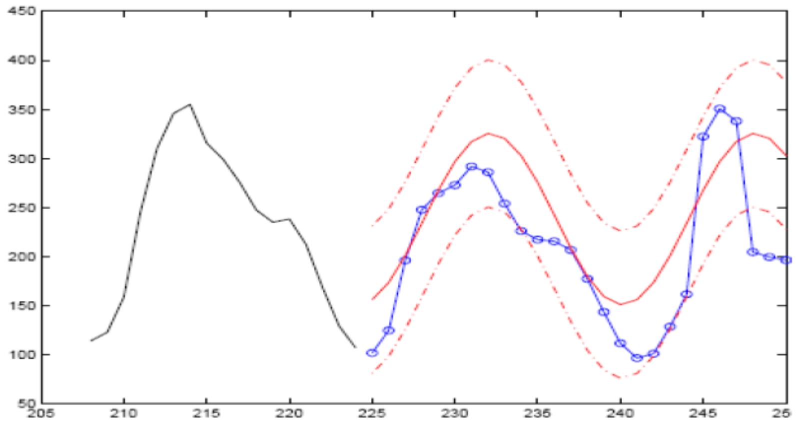
$$MSE_t^2(\ell) = \sigma^2 \left(h_0^2 + \cdots + h_{\ell-1}^2\right) \tag{5.16}$$

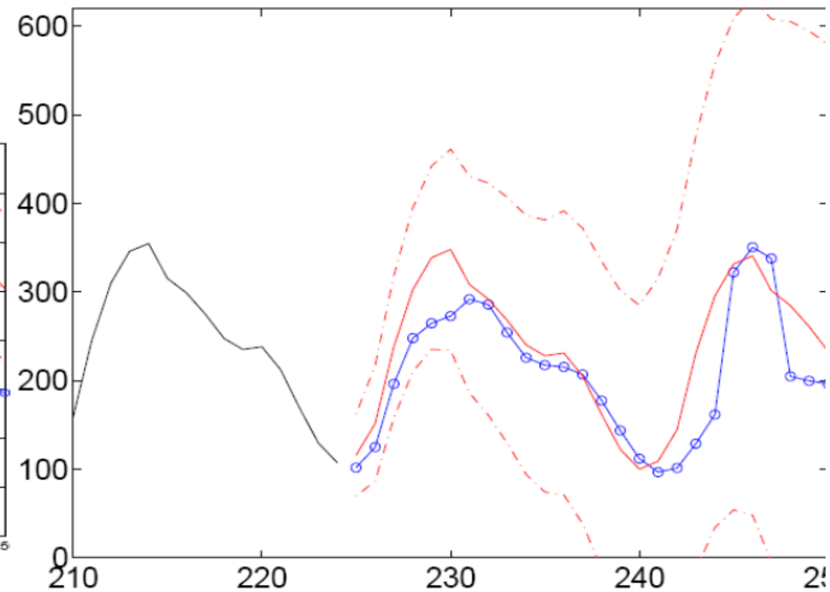
where $h_0, h_1, h_2, \ldots$ *is the impulse response of* L^{-1}. *A prediction interval at level* 0.95 *is thus*

$$\hat{Y}_t(\ell) \pm 1.96\sqrt{MSE_t^2(\ell)} \tag{5.17}$$

Methods Comparison

Compare the Two





Linear Regression with 3 parameters + variance

- Bad point predictions, good prediction intervals
- Need to estimate the paramaters (except from setting a period) here is 16

Assuming differenced data is iid

- Good point predictions, large prediction intervals
- Just need to set the filter lag (similar as setting the period) here is 16